Applicant	:	Fujian GIBO Kitchen&Bath Tech Co., Ltd. Bld. 54, Zone B, Pushang Industrial park, Fuzhou city, 350008, Fujian province, CN.
Sample Description	:	Sensor assembly
Style No.	:	69491
Rated Voltage	:	DC 6V
Test Sample Receipt Date, Location	:	2021-07-22, 2021-08-11, Shenzhen
Test Period, Location	:	From 2021-07-22 to 2021-08-13, Shenzhen
Test Result(s)	:	Refer to Section 3

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Purpose Of Examination / Conclusion:

No.	Test Item(s)	Conclusion
	Restriction of Hazardous Substances (RoHS) Directive 2011/65/EU and its amendment Directive (EU) 2015/863	
1.	Quantitative screening of Lead, Cadmium, Mercury, Chromium and Bromine	Pass
2.	Heavy Metals Content Test (Total Cadmium, Total Lead, Total Mercury, Hexavalent Chromium)	Pass
3.	Brominated Flame Retardants (PBBs & PBDEs)	Pass
4.	Phthalates (DEHP, BBP, DBP and DIBP)	Pass

Remarks:
(1) The results relate only to the items tested.
(2) Samples are tested as received.
(3) "Pass" means the measured result is within a limit, even when extended by expanded uncertainty. "Fail" means the measured result is beyond a limit, even when extended by expanded uncertainty. "Inconclusive" means the measured result can be within or beyond a limit when extended by expanded uncertainty. The confidence level of the expanded uncertainty for "Pass", "Fail" and "Inconclusive" is 95%.
(4) The test sample was specified by the client.

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TUV SUD Group

Prepared by: Carrie

Reviewed by: Eva

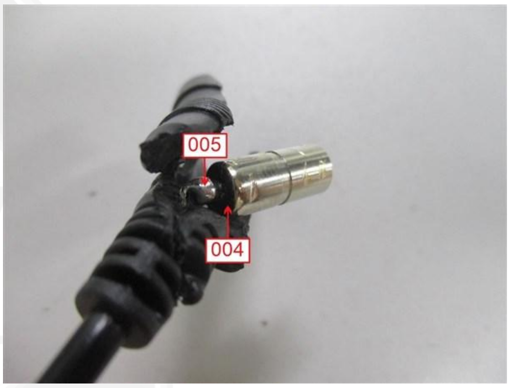
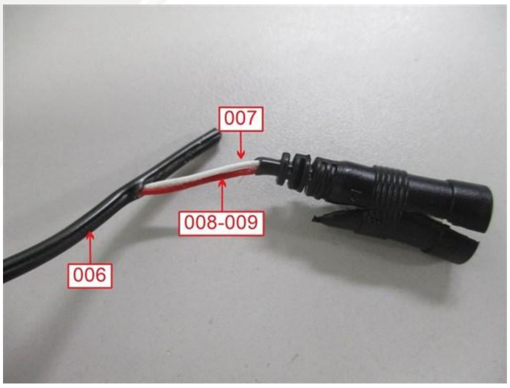
Carrie Liu
Senior Project Coordinator

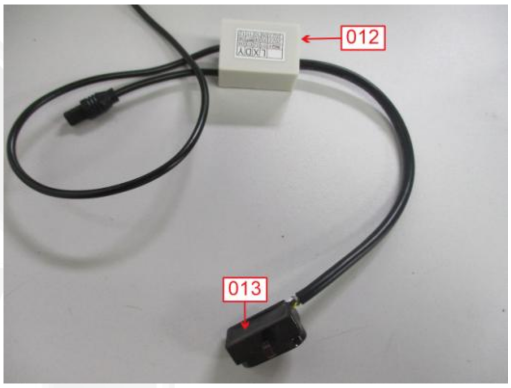
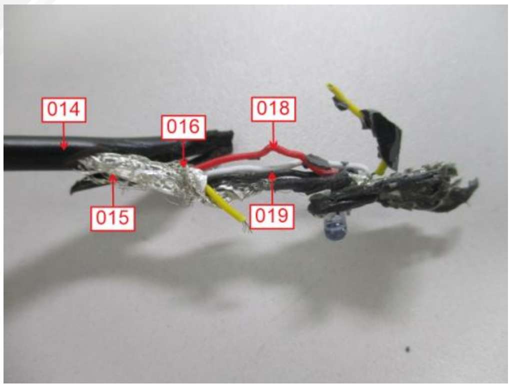
Eva Liang
Project Manager

1. Description of the Test Sample:

Sample Description	Sensor assembly

2. List of Materials as identified by the Laboratory:

T. No.	Sample No.	Colour and Description	Photograph
T1	001	Silvery metal (Pin of connector)	
T2	002	Silvery metal (Connector)	
T3	003	Black soft plastic (Shell of connector)	
T4	004	Black plastic (Terminal of connector)	
T5	005	Silvery metal (Solder of connector)	
T6	006	Black soft plastic (Wire jacket)	
T7	007	White soft plastic (Wire jacket)	
T8	008	Dark red soft plastic (Wire jacket)	
T9	009	Silvery metal (Inner wire jacket)	

T. No.	Sample No.	Colour and Description	Photograph
T10	010	Silvery metal (Pin of small connector)	
T11	012	Grey plastic (Shell of capacitor)	
T12	013	Dark red plastic (Shell of LED)	
T13	014	White printed black soft plastic (Thick wire jacket)	
T14	015	Silvery metal (Film inner wire jacket)	
T15	016	Silvery metal (Iron wire inner wire jacket)	
T16	018	Red soft plastic (Wire jacket)	
T17	019	Black soft plastic (Wire jacket)	

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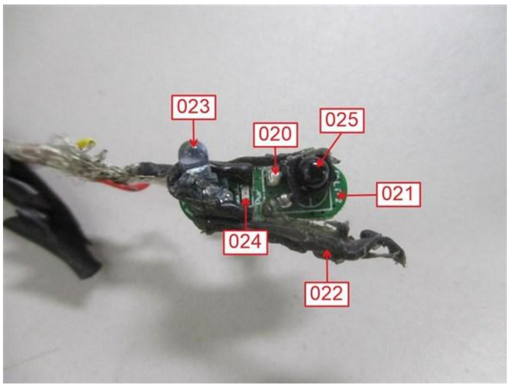
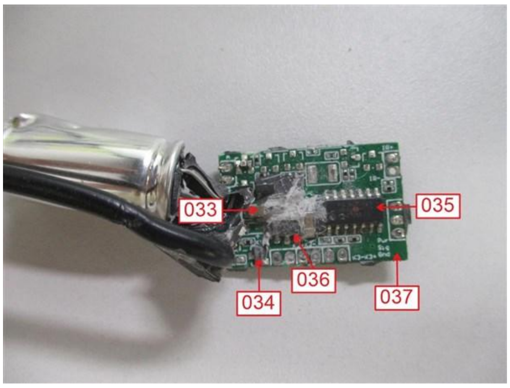
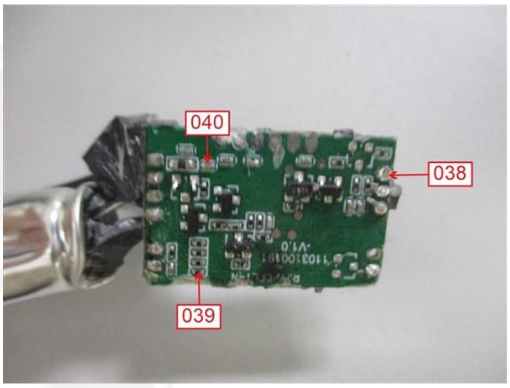
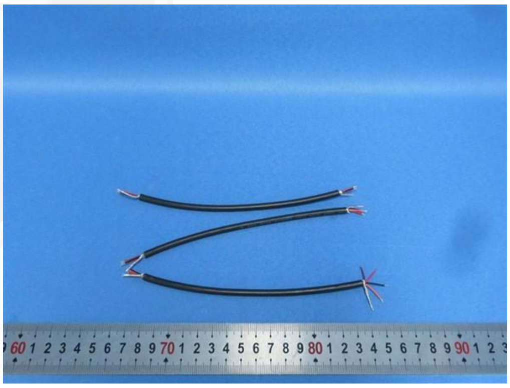
T. No.	Sample No.	Colour and Description	Photograph
T18	020	Silvery metal (Solder of LED PCB)	
T19	021	White printed green PCB (PCB of LED)	
T20	022	Matt black glue (On LED PCB)	
T21	023	Translucent light blue plastic (LED of PCB)	
T22	024	Translucent white material w/ pin (Chip LED of PCB)	
T23	025	Black material w/ pin (Electronic component of LED PCB)	
T24	026	Black glue (In capacitor shell)	
T25	027	White printed black plastic sleeve (Capacitor)	
T26	028	Silvery metal case (Capacitor)	
T27	029	Brown paper (Capacitor)	
T28	030	Silvery metal foil (Capacitor)	
T29	031	Black soft plastic base (Capacitor)	
T30	032	Silvery metal pin (Capacitor)	

T. No.	Sample No.	Colour and Description	Photograph
T31	033	Brown material w/ pin (Chip capacitor of PCB)	
T32	034	Black material w/ pin (Chip triode of PCB)	
T33	035	Black material w/ pin (Chip IC of PCB)	
T34	036	Black material w/ pin (Chip electronic component of PCB)	
T35	037	White printed green PCB (Big PCB)	
T36	038	Silvery metal (Solder of big PCB)	
T37	039	Black material w/ pin (Small chip resistor of PCB)	
T38	040	Brown material w/ pin (Small chip capacitor of PCB)	
T39	041	Brown soft plastic (Wire jacket)	
T40	042	Orange soft plastic (Wire jacket)	

3. Test Result

3.1 Quantitative screening of Lead, Cadmium, Mercury, Chromium and Bromine

Test Method: With reference to EN 62321-3-1:2014, analyzed by Energy-Dispersive X-Ray Fluorescence Spectrometers (XRF)
[Reporting Limit = 50 mg/kg for Cadmium, 300 mg/kg for Bromine, and 500 mg/kg for others]

Sample	Total Cadmium [mg/kg]	Total Lead [mg/kg]	Total Chromium [mg/kg]	Total Mercury [mg/kg]	Total Bromine [mg/kg]
001	BL	**OL**(a)	BL	BL	NA
002	BL	**OL**(a)	BL	BL	NA
004	BL	BL	BL	BL	BL
005	BL	BL	BL	BL	NA
009	BL	BL	BL	BL	NA
010	BL	BL	BL	BL	NA
012	BL	BL	BL	BL	BL
013	BL	BL	BL	BL	BL
015	BL	BL	BL	BL	NA
016	BL	BL	BL	BL	NA
018	BL	BL	BL	BL	BL
019	BL	BL	BL	BL	BL
020	BL	BL	BL	BL	NA
021	BL	BL	BL	BL	**Inconclusive**(b)
RoHS Requirement	100	1000	1000	1000	--

Note 1. "mg/kg" denotes milligram per kilogram
2. "BL" denotes below the limit
3. "OL" denotes over the limit
4. "N/A" denotes Not Applicable
5. "(a)" denotes further confirmation test was conducted, results are listed in 3.2
6. "(b)" denotes further confirmation test was conducted, results are listed in 3.3

3.1 Quantitative screening of Lead, Cadmium, Mercury, Chromium and Bromine

Test Method: With reference to EN 62321-3-1:2014, analyzed by Energy-Dispersive X-Ray Fluorescence Spectrometers (XRF)
[Reporting Limit = 50 mg/kg for Cadmium, 300 mg/kg for Bromine, and 500 mg/kg for others]

Sample	Total Cadmium [mg/kg]	Total Lead [mg/kg]	Total Chromium [mg/kg]	Total Mercury [mg/kg]	Total Bromine [mg/kg]
022	BL	BL	BL	BL	BL
023	BL	BL	BL	BL	BL
024	BL	BL	BL	BL	BL
025	BL	BL	BL	BL	BL
026	BL	BL	BL	BL	BL
027	**OL**(a)	BL	BL	BL	BL
028	BL	BL	BL	BL	NA
029	BL	BL	BL	BL	BL
030	BL	BL	BL	BL	NA
031	BL	BL	BL	BL	BL
032	BL	BL	BL	BL	NA
033	BL	BL	BL	BL	BL
034	BL	BL	BL	BL	**Inconclusive**(b)
035	BL	BL	BL	BL	BL
036	BL	BL	BL	BL	BL
037	BL	BL	BL	BL	BL
038	BL	BL	BL	BL	NA
039	BL	BL	**Inconclusive**(a)	BL	BL
040	**OL**(a)	BL	BL	BL	BL
RoHS Requirement	100	1000	1000	1000	--

Note 1. “mg/kg” denotes milligram per kilogram
2. “BL” denotes below the limit
3. “OL” denotes over the limit
4. “N/A” denotes Not Applicable
5. “(a)” denotes further confirmation test was conducted, results are listed in 3.2
6. “(b)” denotes further confirmation test was conducted, results are listed in 3.3

XRF screening limits in mg/kg for regulated elements in various matrices

ELEMENT	POLYMER		
	BL	INCONCLUSIVE	OL
Cd	X<(70-3σ)	(70-3σ)<X<(130+3σ)	X>(130+3σ)
Pb	X<(700-3σ)	(700-3σ)<X<(1300+3σ)	X>(1300+3σ)
Hg	X<(700-3σ)	(700-3σ)<X<(1300+3σ)	X>(1300+3σ)
Br	X<(300-3σ)	X>(300-3σ)	NA
Cr	X<(700-3σ)	X>(700-3σ)	NA

ELEMENT	METAL		
	BL	INCONCLUSIVE	OL

Cd	X<(70-3σ)	(70-3σ)<X<(130+3σ)	X>(130+3σ)
Pb	X<(700-3σ)	(700-3σ)<X<(1300+3σ)	X>(1300+3σ)
Hg	X<(700-3σ)	(700-3σ)<X<(1300+3σ)	X>(1300+3σ)
Cr	X<(700-3σ)	X>(700-3σ)	NA

ELEMENT	COMPLEX MATERIAL		
	BL	INCONCLUSIVE	OL
Cd	X<(50-3σ)	(50-3σ)<X<(150+3σ)	X>(150+3σ)
Pb	X<(500-3σ)	(500-3σ)<X<(1500+3σ)	X>(1500+3σ)
Hg	X<(500-3σ)	(500-3σ)<X<(1500+3σ)	X>(1500+3σ)
Br	X<(250-3σ)	X>(250-3σ)	NA
Cr	X<(500-3σ)	X>(500-3σ)	NA

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3.2 Heavy Metals Content Test (Total Cadmium, Total Lead, Total Mercury, Hexavalent Chromium)

Test method: with reference to EN 62321-4:2014+A1:2017, EN 62321-5:2014,
EN 62321-7-1:2015 and EN 62321-7-2:2017 (For Hexavalent Chromium), analyzed by Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES) and UV-Vis spectrophotometer.
[Reporting Limit= 10.0 mg/kg for Cadmium, Lead, Mercury and Chromium VI,
0.1 µg/cm² for Chromium VI for metal]

Sample	Total Cadmium [mg/kg]	Total Lead [mg/kg]	Total Mercury [mg/kg]	Hexavalent Chromium [mg/kg]	Conclusion
001	--	**29070.7**	--	--	**Pass(c)**
002	--	**28915.6**	--	--	**Pass(c)**
003	<2.0	<10.0	<10.0	<10.0	Pass
006	<2.0	<10.0	<10.0	<10.0	Pass
007	<2.0	<10.0	<10.0	<10.0	Pass
008	<2.0	<10.0	<10.0	<10.0	Pass
014	<2.0	<10.0	<10.0	<10.0	Pass
027	<2.0	--	--	--	Pass
039	--	--	--	<10.0	Pass
040	<2.0	--	--	--	Pass
041	<2.0	<10.0	<10.0	<10.0	Pass
042	<2.0	<10.0	<10.0	<10.0	Pass
RoHS Requirement	100	1000	1000	1000	--

Note 1. "< " denotes less than.
2. "mg/kg" denotes milligram per kilogram.
3. "--" denotes tested by XRF, result is listed in Section 3.1
4. -"(c)" denotes the exempt item according to DIRECTIVE 2011/65/EU Annex III item 6(c) "Copper alloy containing up to 4 % lead by weight".

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3.3 Brominated Flame Retardants (PBBs & PBDEs)

Test Method: With reference to EN 62321-6:2015, extraction with toluene and analyzed by Gas Chromatography and Mass Spectrometry (GC-MS).
[Reporting Limit = 5mg/kg]

Test Item		Results [mg/kg]			RoHS Requirement [mg/kg]
		Sample 003	Sample 006	Sample 007	
PBBs	Monobromobiphenyl	<5	<5	<5	Sum of PBBs < 1000
	Dibromobiphenyl	<5	<5	<5	
	Tribromobiphenyl	<5	<5	<5	
	Tetrabromobiphenyl	<5	<5	<5	
	Pentabromobiphenyl	<5	<5	<5	
	Hexabromobiphenyl	<5	<5	<5	
	Heptabromobiphenyl	<5	<5	<5	
	Octabromobiphenyl	<5	<5	<5	
	Nonabromobiphenyl	<5	<5	<5	
	Decabromobiphenyl	<5	<5	<5	
	Sum of detected PBBs	<50	<50	<50	
PBDEs	Monobromodiphenyl ether	<5	<5	<5	Sum of PBDEs < 1000
	Dibromodiphenyl ether	<5	<5	<5	
	Tribromodiphenyl ether	<5	<5	<5	
	Tetrabromodiphenyl ether	<5	<5	<5	
	Pentabromodiphenyl ether	<5	<5	<5	
	Hexabromodiphenyl ether	<5	<5	<5	
	Heptabromodiphenyl ether	<5	<5	<5	
	Octabromodiphenyl ether	<5	<5	<5	
	Nonabromodiphenyl ether	<5	<5	<5	
	Decabromodiphenyl ether	<5	<5	<5	
	Sum of detected PBDEs	<50	<50	<50	

Note 1. “<” denotes less than
2. “mg/kg” denotes milligram per kilogram

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3.3 Brominated Flame Retardants (PBBs & PBDEs)

Test Method: With reference to EN 62321-6:2015, extraction with toluene and analyzed by Gas Chromatography and Mass Spectrometry (GC-MS).
[Reporting Limit = 5mg/kg]

Test Item		Results [mg/kg]			RoHS Requirement [mg/kg]
		Sample 008	Sample 014	Sample 021	
PBBs	Monobromobiphenyl	<5	<5	<5	Sum of PBBs < 1000
	Dibromobiphenyl	<5	<5	<5	
	Tribromobiphenyl	<5	<5	<5	
	Tetrabromobiphenyl	<5	<5	<5	
	Pentabromobiphenyl	<5	<5	<5	
	Hexabromobiphenyl	<5	<5	<5	
	Heptabromobiphenyl	<5	<5	<5	
	Octabromobiphenyl	<5	<5	<5	
	Nonabromobiphenyl	<5	<5	<5	
	Decabromobiphenyl	<5	<5	<5	
	Sum of detected PBBs	<50	<50	<50	
PBDEs	Monobromodiphenyl ether	<5	<5	<5	Sum of PBDEs < 1000
	Dibromodiphenyl ether	<5	<5	<5	
	Tribromodiphenyl ether	<5	<5	<5	
	Tetrabromodiphenyl ether	<5	<5	<5	
	Pentabromodiphenyl ether	<5	<5	<5	
	Hexabromodiphenyl ether	<5	<5	<5	
	Heptabromodiphenyl ether	<5	<5	<5	
	Octabromodiphenyl ether	<5	<5	<5	
	Nonabromodiphenyl ether	<5	<5	<5	
	Decabromodiphenyl ether	<5	<5	<5	
	Sum of detected PBDEs	<50	<50	<50	

Note 1. “<” denotes less than
2. “mg/kg” denotes milligram per kilogram

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3.3 Brominated Flame Retardants (PBBs & PBDEs)

Test Method: With reference to EN 62321-6:2015, extraction with toluene and analyzed by Gas Chromatography and Mass Spectrometry (GC-MS).
[Reporting Limit = 5mg/kg]

Test Item		Results [mg/kg]			RoHS Requirement [mg/kg]
		Sample 034	Sample 041	Sample 042	
PBBs	Monobromobiphenyl	<5	<5	<5	Sum of PBBs < 1000
	Dibromobiphenyl	<5	<5	<5	
	Tribromobiphenyl	<5	<5	<5	
	Tetrabromobiphenyl	<5	<5	<5	
	Pentabromobiphenyl	<5	<5	<5	
	Hexabromobiphenyl	<5	<5	<5	
	Heptabromobiphenyl	<5	<5	<5	
	Octabromobiphenyl	<5	<5	<5	
	Nonabromobiphenyl	<5	<5	<5	
	Decabromobiphenyl	<5	<5	<5	
	Sum of detected PBBs	<50	<50	<50	
PBDEs	Monobromodiphenyl ether	<5	<5	<5	Sum of PBDEs < 1000
	Dibromodiphenyl ether	<5	<5	<5	
	Tribromodiphenyl ether	<5	<5	<5	
	Tetrabromodiphenyl ether	<5	<5	<5	
	Pentabromodiphenyl ether	<5	<5	<5	
	Hexabromodiphenyl ether	<5	<5	<5	
	Heptabromodiphenyl ether	<5	<5	<5	
	Octabromodiphenyl ether	<5	<5	<5	
	Nonabromodiphenyl ether	<5	<5	<5	
	Decabromodiphenyl ether	<5	<5	<5	
	Sum of detected PBDEs	<50	<50	<50	

Note 1. “<” denotes less than
2. “mg/kg” denotes milligram per kilogram

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3.4 Phthalates (DEHP, BBP, DBP and DIBP)

Test Method: with reference to EN 62321-8:2017, extracted by organic solvent and analyzed by Gas Chromatography and Mass Spectrometry (GC-MS).
[Reporting Limit = 50mg/kg]

Test Item	CAS No.	Result [mg/kg]		
		Sample 003+006+007	Sample 004+012+013	Sample 008+014+041
Di-iso-butyl Phthalate (DIBP)	84-69-5	<50	<50	<50
Di-butyl Phthalate (DBP)	84-74-2	<50	<50	<50
Butyl-benzyl Phthalate (BBP)	85-68-7	<50	<50	<50
Di-(2-ethyl-hexyl) Phthalate (DEHP)	117-81-7	<50	<50	<50
RoHS Requirement		Each < 1000		

Test Item	CAS No.	Result [mg/kg]		
		Sample 018+019+031	Sample 021+023	Sample 022+026
Di-iso-butyl Phthalate (DIBP)	84-69-5	<50	<50	<50
Di-butyl Phthalate (DBP)	84-74-2	<50	<50	<50
Butyl-benzyl Phthalate (BBP)	85-68-7	<50	<50	<50
Di-(2-ethyl-hexyl) Phthalate (DEHP)	117-81-7	<50	<50	<50
RoHS Requirement		Each < 1000		

Note 1. “<” denotes less than
2. “mg/kg” denotes milligram per kilogram

3.4 Phthalates (DEHP, BBP, DBP and DIBP)

Test Method: with reference to EN 62321-8:2017, extracted by organic solvent and analyzed by Gas Chromatography and Mass Spectrometry (GC-MS).
[Reporting Limit = 50mg/kg]

Test Item	**CAS No.**	**Result [mg/kg]**	
		Sample 027+037	**Sample 042**
Di-iso-butyl Phthalate (DIBP)	84-69-5	<50	<50
Di-butyl Phthalate (DBP)	84-74-2	<50	<50
Butyl-benzyl Phthalate (BBP)	85-68-7	<50	<50
Di-(2-ethyl-hexyl) Phthalate (DEHP)	117-81-7	<50	<50
RoHS Requirement		Each < 1000	

Note 1. “<” denotes less than
2. “mg/kg” denotes milligram per kilogram

-- END OF THE TEST REPORT --